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Automatic Certificate Generation Using Python

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Automation projects are nowadays highly appreciated due to the availability of cloud computing resources. This article is in continuation to the ‘Automatic Certificate Generation Using MATLAB’ DIY article published earlier and also available at *https://www.electronicsforu.com/electronics-projects/automatic-certificate-generation-matlab*, which garnered much attention because of the code’s readability and scalability. Several academicians, especially students, contacted us regarding the project and its future scope. This motivated us to create a similar project using Python.

Python will help to overcome a few challenges in MATLAB. The project uses only online tools and packages, hence does not require any installation. Python language is used in all applications ranging from image processing to artificial intelligence. Fig. 1 represents the basic idea about generating certificates using Python.

The Python script is run from an online tool called Colab. which accesses a sheet containing information, an image file that is a blank certificate and a ttf- font file to decide font type. When the script is run, a zipped folder is generated that can be downloaded to the local system from the browser. The folder contains all the certificates generated by the project. The code can be easily modified for generating any number of certificates and any number of fields to be printed on the certificates.

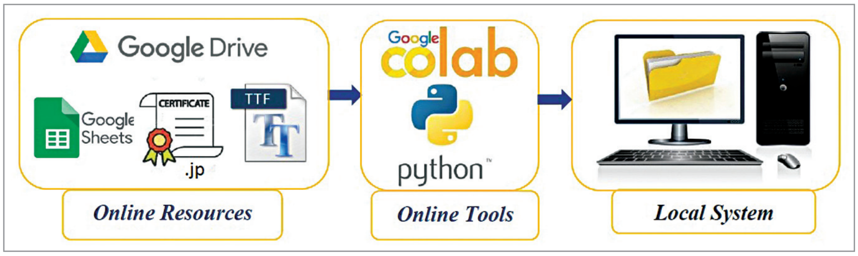

Fig. 1: Block diagram of the project

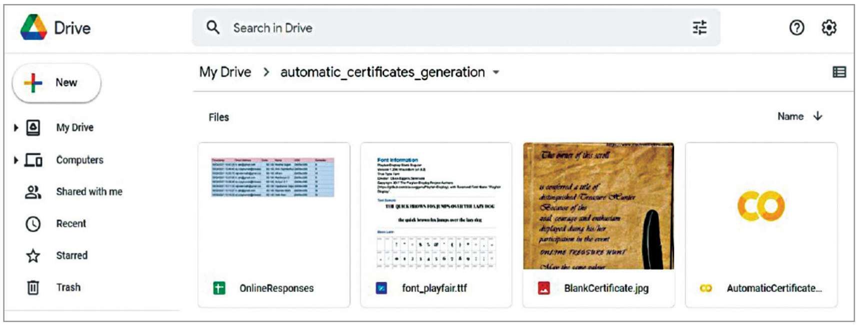

Fig. 2: Work environment

Fig. 3: Initiating Colab

Fig. 4: Running the code in Colab

Advantages of Python on Colab

The project uses Colaboratory, colab in short—a Google cloud based platform that provides a Jupyter notebook-like environment to write and execute Python code on the go. Colab offers some of the free services to the users, which are sufficient to carry out minor projects involving data science, artificial intelligence, machine learning, and deep learning. Apart from providing free runtime with around 12Gb RAM, it provides access to its GPUs and TPUs that can accelerate the training process when working with huge datasets.

Colab has almost all the standard libraries pre-installed, offering users the plug-and-play functionality that saves them from the headache of library dependency. Moreover, the files generated while executing

Requirements

Google Account (Gmail), Internet Browser, Internet and unzipping software (optional). Unzipping software is required to open and access the certificates which are in the zipped folder.